CHECK-IN 2 BEFORE END-SEM · COMPUTER VISION AND PERCEPTION

Right in meaning, wrong in place?

ASP asks a language model to guess the rooms and objects a robot has not yet seen, then drives to where the guesses disagree. Its author says the guesses are often in the wrong place but name the right things, and that this is enough. We test that claim.

01

Last check-in

Does a better guesser make a better explorer? Finish the live test, screen models, re-test our filter.

02

Since then

The live test finished. We rebuilt the paper's reference and scored every run with the author's own code.

03

End-sem

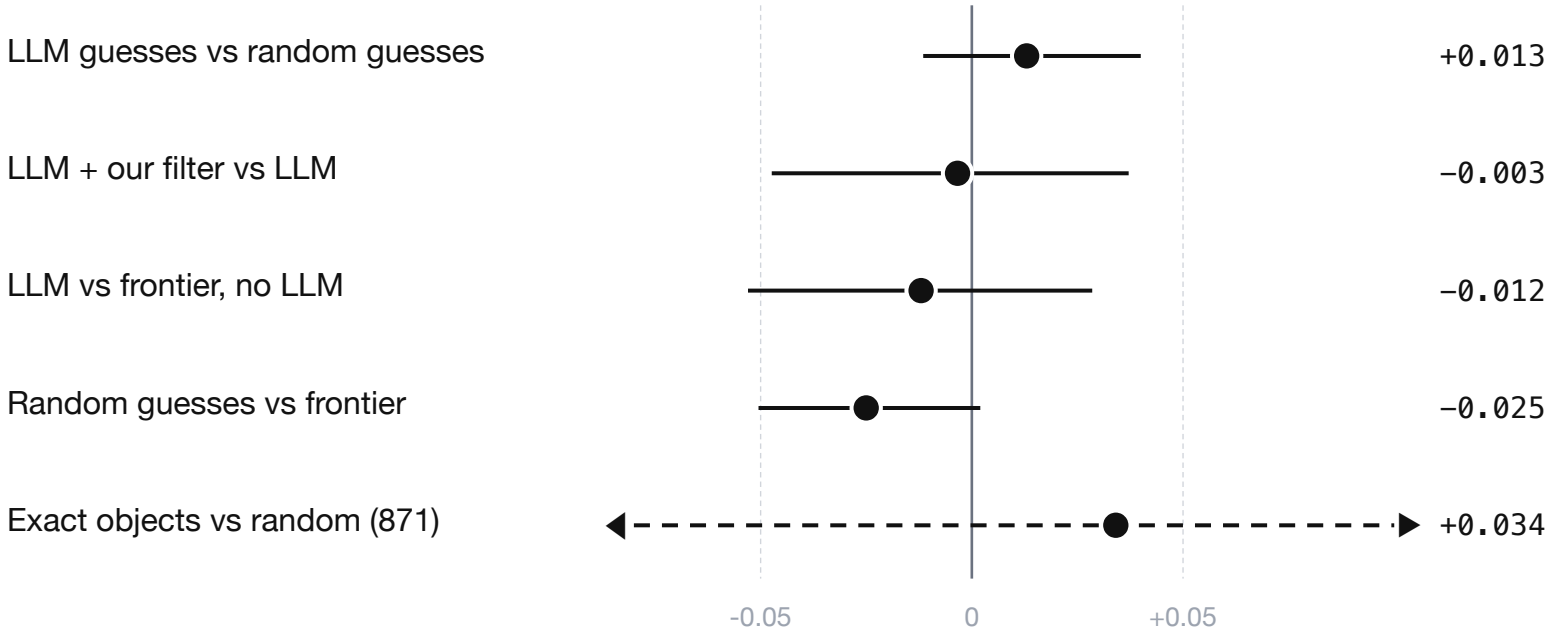
A pre-registered test of the author's defense, with ASP scored on the paper's own scale.

Sher Partap Singh · Preesha Katial · Ananya Singla

Base paper: Tang & Chaudhari, Active Semantic Perception, arXiv:2510.05430

Live, no gain over random guesses that we could detect

DIFFERENCE IN MAPPING QUALITY (F1 AREA TO 60 M), WITH ITS 95% INTERVAL



Registered tests, pilot status. Each interval crosses zero. Exact objects: 871 only, 3 runs v 3; all 3 beat all 3 random runs, but p 0.10 is the floor and the interval is unbounded. The LLM arm has 2 runs on 853. A secondary test, F1 at 60 m, favoured the LLM (+0.043, p 0.015; 0.09 after correction).

WHERE THE LLM PUTS NEW OBJECTS

3.0%	0.5%
LLM guesses	random guesses

New objects within 0.5 m of a real object with the same name, all counted live runs (not tested). 76% of the LLM's are more than 2 m off.

Correction: last time we showed 1.7% vs 1.5%; on the 8 matched saved inputs it is 2.1% vs 1.4%.

Reading. The LLM lands near a real object 6× as often as random guesses, but 97% miss by over 0.5 m, and we could not detect a better map.

On the paper's scale, his 853 ASP lead does not appear

NOW AS IN THE PAPER

Scoring. the author's own F1 and GED code, on his time axis.

Reference. his protocol, scripted: a tour of every reachable point on the start floor, same mapper (85 and 66 stops). It holds 32% and 21% of the human-labelled objects; its repeat tour failed.

Frontier. run to its natural end: 5 of 5 runs on 853, 1 of 5 on 871.

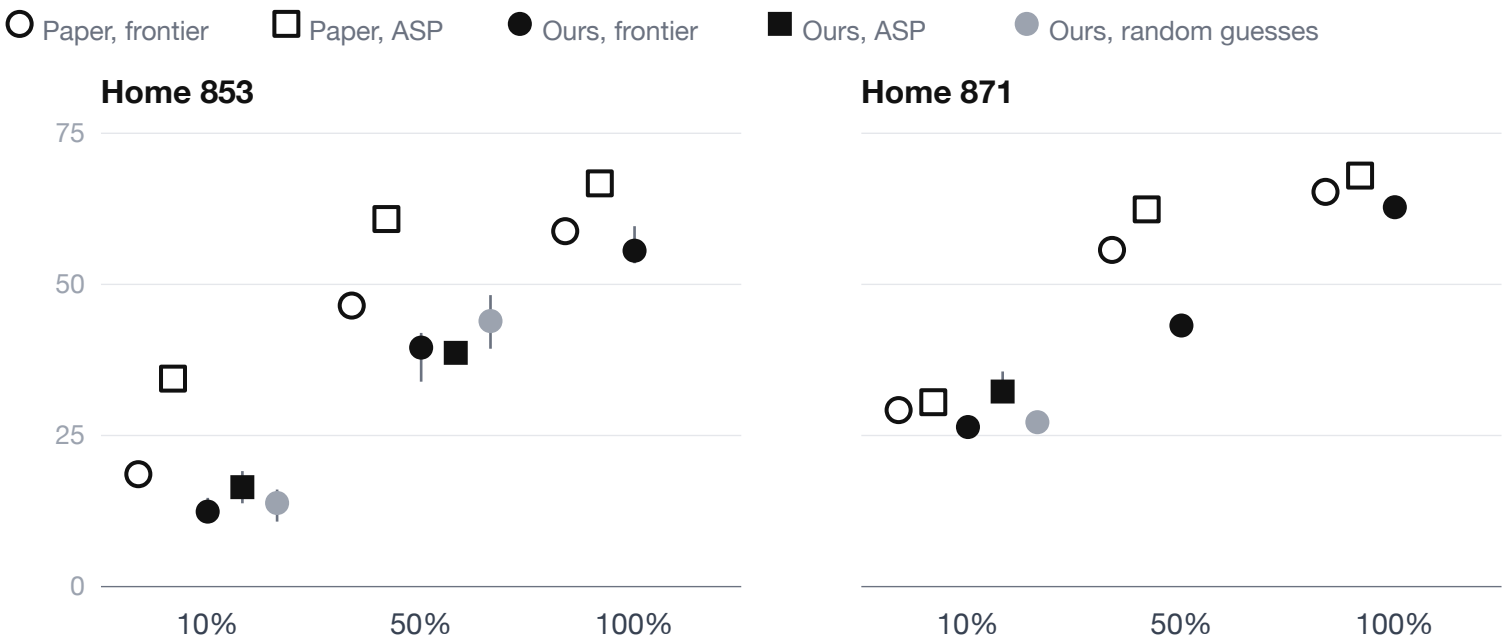
WHERE WE DIFFER

Budget. the paper gives none; we use our frontier's median end, 632 s (853), and the registered fallback, 809 s (871).

LLM. ours DeepSeek, his Gemini 2.5 Pro.

871: 4 frontier runs crashed twice when another user's server took the GPU. A third try waits for a free GPU.

OBJECT F1 × 100 AT 10, 50 AND 100% OF THE BUDGET



Hollow: the paper's single run against its own reference (Fig. 4 left). Filled: our mean against the rebuilt reference, bars the range; shown where at least half the runs reach that time (our rule). 853 at 100%: 3 of 5 runs. Our ASP runs stop at 60 m and ran in October; frontier ran in November.

Reading. Our frontier ends within about 3 points of his. On 853 his ASP leads frontier by 14 points at 50%; our 2 ASP runs are level (38.7 vs 39.5), and random guesses score 43.9.

Next: is the meaning right, and would it be enough?

01

Right beyond base rates?

Do the rooms and objects named really lie in unexplored space? Compare with controls that see the same map: house statistics, another house, shuffled guesses.

WHY · The prompt says a scene “may contain more than one bedroom”, so naming one proves little.

02

Would it be enough?

Offline, give ASP exact labels with the placement error real models make, and see how much of the oracle's ranking power survives.

WHY · ASP rewards disagreement between guesses, so guesses that all agree add nothing to its view score.

03

Is it the model?

Replay Claude on the same 44 saved inputs as DeepSeek, so a gap points to the model, not the inputs.

WHY · About $320 expected, capped at $460; we ask before spending.

04

Our validity layer, same pass

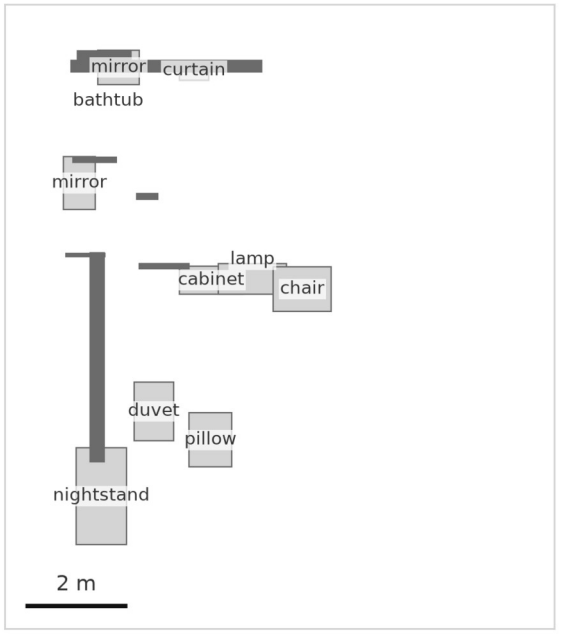
For every model: what the filter removes, and whether what it keeps is right more often.

WHY · Live, the filter made no difference we could detect (–0.003).

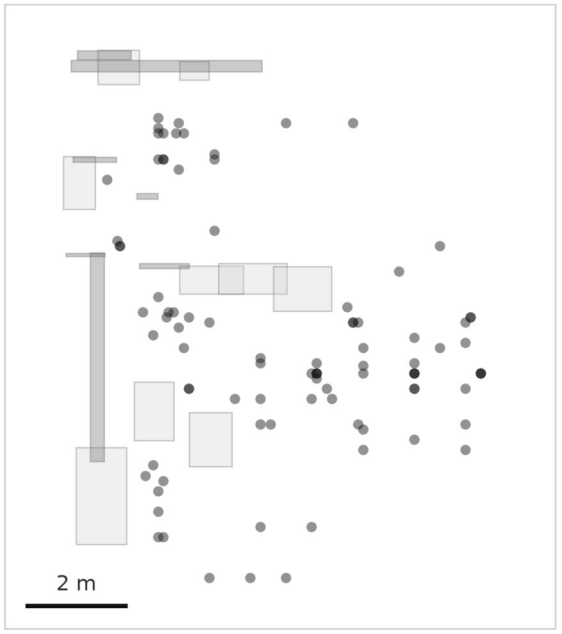
Each step is $0 except Claude. We hash the rules before any new score. DeepSeek results stay exploratory, because we saw pilot numbers first; the Claude replays are the blind test.

What end-sem will test, in one stage of one run

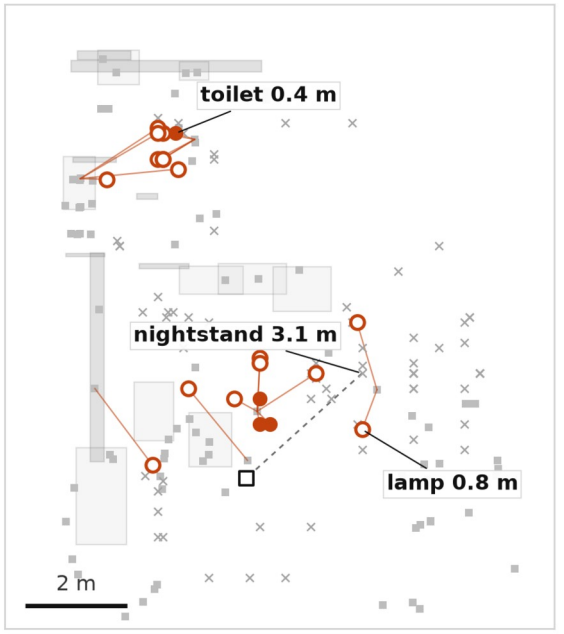
1 · MAPPED SO FAR



2 · GUESSED, ALL 8 MEMBERS



3 · REAL OBJECTS, MATCHED



DeepSeek, run e1_00853_s42_official_60m, stage 3, 8 members, home 853. Top-down, z 0.5 to 1.5 m, habitat frame. A new guess lies more than 0.5 m from every mapped object. 46 of 90 have a name no real object in the house has.

HOW TO READ PANEL 3

- Same-name real object within 0.5 m
- Within 2 m
- × Farther, or no such object
- Real object (human label)

A label gives the distance to the nearest real object with that name. Darker marks in panel 2: members agree.

90	new guesses
4	within 0.5 m of a real one
20	within 2 m

All 15 sample stages: 30 of 1230 within 0.5 m, 307 within 2 m.

End-sem question. Are the names right beyond base rates, even where the places are wrong? We score this on the same stages, with the validity layer and the Claude replays. Result: a pre-registered test of “right in meaning, wrong in place?”.